Preso

Good morning, I will present the law of total progeny (W) this morning. First, let's recall some notation that we will need. The intermediate and the main result I will discuss all use the same notation.

- W denotes the total progeny of a Galton–Watson branching process started from one ancestor
- let $A_t = A_{t-1} + X_t - 1$ with $A_0 = 1$, $t \geq 1$, be the breadth-first queue process, where the offspring variables $X_1, X_2, \ldots$ are i.i.d. with common offspring distribution.
- The random walk representation of the GW tree is $S_t = 1 + \sum_{i=1}^{t} Y_i$ where $Y_i = X_i - 1$. Here, S_t tracks the queue in the GW exploration, so $A_t = S_t$.
- The time the queue process empties is $\tau_0 := \inf\{t \geq 0 : A_t = 0\} = \inf\{t \geq 0 : S_t = 0\}$.

Now, I am going to state two results without proof as facts. The proofs are in the notes and the textbook by Sebastian Roch.

The first is Lemma 1, which is Lemma 6.2.4 in the Roch book. It states that total progeny is the extinction time of the queue/exploration process A_t: $W = \tau_0$, where τ_0 is the first time the queue process hits 0.

The second is Theorem 4, which is Theorem 6.2.5 in the Roch book, and is known as the hitting time theorem. Suppose we have a random walk $(R_t)_{t\geq 0}$ with $R_0 = 0$ and iid increment process $(U_t)_{t\geq 0}$ such that $P(U_t \leq 1) = 1$. Let $\sigma_\ell := \inf\{t \geq 0 : R_t = \ell\}$ to be the first hitting time of ℓ by process R_t. Then, The theorem says that the probability of first hitting level l at time t is l/t times the probability of simply being at level l at time t., $P(\sigma_\ell = t) = \frac{\ell}{t} P(R_t = \ell)$.

Now, the theorem that I will state and prove is Theorem 5 in the notes, and Theorem 6.2.6 in the Roch book, and is called the Law of Total Progeny. It is an application of Theorem 4 to GW processes, and expresses the probability P(W=t) in terms of the probability that the sum of iid RVs equals t-1, up to a factor 1/t.

Theorem (Law of Total Progeny). For every $t \geq 1$, $\mathbb{P}(W = t) = \frac{1}{t}\,\mathbb{P}(X_1 + \cdots + X_t = t - 1)$.

Proof. Define $\tau_0 := \inf\{t \geq 0 : A_t = 0\}$. By the queue-process lemma, the total progeny equals the first time the queue empties, so $W = \tau_0$. Hence, $\{W = t\} = \{\tau_0 = t\}$.

To apply the hitting-time theorem, we now construct a walk satisfying its assumptions. Now to apply Theorem 4, define a new random walk by $R_t := 1 - S_t$, $t \geq 0$. Then $R_0 = 1 - S_0 = 1 - 1 = 0$. Its increments are $U_t := R_t - R_{t-1} = -(S_t - S_{t-1}) = -(X_t - 1) = 1 - X_t$. Since $X_t \geq 0$, we have $U_t = 1 - X_t \leq 1$ **almost surely**. Moreover, the variables U_t are i.i.d. because the variables X_t are i.i.d.

Next, observe that $S_t = 0 \iff R_t = 1$. Therefore, if we let $\sigma_1 := \inf\{t \geq 0 : R_t = 1\}$, then $\tau_0 = \sigma_1$. So $\mathbb{P}(W = t) = \mathbb{P}(\tau_0 = t) = \mathbb{P}(\sigma_1 = t)$.

We may now apply the hitting-time theorem to the walk R_t with level $\ell = 1$. This gives $\mathbb{P}(\sigma_1 = t) = \frac{1}{t}\,\mathbb{P}(R_t = 1)$. Hence, $\mathbb{P}(W = t) = \frac{1}{t}\,\mathbb{P}(R_t = 1)$.

Finally, since $R_t = 1 - S_t$, the event $\{R_t = 1\}$ is the same as $\{S_t = 0\}$. Also, $S_t = 1 + \sum_{i=1}^{t}(X_i - 1) = 1 + \sum_{i=1}^{t} X_i - t = \sum_{i=1}^{t} X_i - (t-1)$. Therefore,$S_t = 0 \iff \sum_{i=1}^{t} X_i = t-1$. Substituting this into the previous formula yields $\mathbb{P}(W = t) = \frac{1}{t}\,\mathbb{P}\big(X_1 + \cdots + X_t = t - 1\big)$. This proves the result. $\square$

Next, for a simple example of the theorem, we consider a Poisson branching process. Recall that Poisson distribution has one parameter denoted $\lambda > 0$ and its pmf is given by $P(X = k) = p_k = \lambda^k/k!\exp(-\lambda)$ and has the property that the λ is also the mean and variance $\lambda = E[X] = Var[X]$ for a random variable X with the Poisson distribution. Another useful property is that the sum of Poisson random variables $X_i \sim Pois(\lambda)$ is also a Poisson random variable with parameter equal to the sum of λ, i.e., $\sum_{i=1}^{n} X_i \sim Pois(n\lambda)$. Now, consider a GW branching process where the offspring distribution is Poisson with parameter λ, $X_i \sim Pois(\lambda)$. Then, the distribution of total progeny is given by $P(W = t) = 1/tP(X_1 + \cdots X_t = t - 1) = 1/t\lambda^(t - 1)/(t - 1)!\exp(-t\lambda)$ for $t \geq 1$. As you can see, the probability mass of total progeny can be very easily computed using the theorem; otherwise, directly computing W would be much more difficult.

0.1 Script

Good morning. Today I will present the law of total progeny W for a Galton–Watson branching process.

First, let me recall the notation we need.

- W denotes the total progeny of a GW BP started from one ancestor.
- $(\mathcal{A}_t)_{t\geq 0}$ with $\mathcal{A}_0 = \{0\}$ is the BF queue process, and $A_t := |\mathcal{A}_t| = A_{t-1} + X_t - 1$ where the offspring variables $X_1, X_2, \ldots$ are i.i.d. with the common offspring distribution.
- We also define the random walk representation $S_t = 1 + \sum_{i=1}^{t} Y_i$, $Y_i = X_i - 1$. This walk tracks the queue in the BF exploration, so $A_t = S_t$.
- Finally, let $\tau_0 := \inf\{t \geq 0 : A_t = 0\} = \inf\{t \geq 0 : S_t = 0\}$, the first time the queue process empties.

Next, I will state two results without proof as facts. The proofs are in the notes and the Roch book.

First, Lemma 1 says that the total progeny equals the first time the queue empties: $W = \tau_0$. This is because exactly one individual is explored at each step, so the number of explored steps until the queue empties is the total number of individuals.

Second, Theorem 4, the hitting-time theorem, says the following. If $(R_t)_{t\geq 0}$ is a random walk with $R_0 = 0$, i.i.d. increments (U_t), and $U_t \leq 1$ a.s., and if $\sigma_\ell := \inf\{t \geq 0 : R_t = \ell\}, \ell \in \mathbb{Z}^{++}$ then $\mathbb{P}(\sigma_\ell = t) = \frac{\ell}{t}\mathbb{P}(R_t = \ell)$. We saw a special case of this for SSRW in Exercise 8 in the Week 1 lecture notes, where $U_t \in \{-1, +1\}$ with p=1/2.

Now I state the main result.

Theorem (Law of Total Progeny). For every $t \geq 1$, $\mathbb{P}(W = t) = \frac{1}{t}\,\mathbb{P}(X_1 + \cdots + X_t = t - 1)$.

Proof. By the lemma, $W = \tau_0$, so $\{W = t\} = \{\tau_0 = t\}$.

Next, define a new walk by $R_t := 1 - S_t$. Then $R_0 = 0$, and its increments are $U_t = R_t - R_{t-1} = 1 - X_t$. Since $X_t \geq 0$, we have $U_t \leq 1$ almost surely, and increments are i.i.d.

Also, $S_t = 0 \iff R_t = 1$. So if $\sigma_1 := \inf\{t \geq 0 : R_t = 1\}$, then $\tau_0 = \sigma_1$. Therefore $\mathbb{P}(W = t) = \mathbb{P}(\tau_0 = t) = \mathbb{P}(\sigma_1 = t)$.

Now we apply the hitting-time theorem with $\ell = 1$: $\mathbb{P}(\sigma_1 = t) = \frac{1}{t}\mathbb{P}(R_t = 1)$. Hence $\mathbb{P}(W = t) = \frac{1}{t}\mathbb{P}(R_t = 1)$.

Finally, since $R_t = 1 - S_t$, the event $\{R_t = 1\}$ is the same as $\{S_t = 0\}$. But $S_t = 1 + \sum_{i=1}^{t}(X_i - 1) = \sum_{i=1}^{t} X_i - (t-1)$. So $S_t = 0 \iff \sum_{i=1}^{t} X_i = t - 1$. Substituting this gives $\mathbb{P}(W = t) = \frac{1}{t}\mathbb{P}(X_1 + \cdots + X_t = t - 1)$. This proves the theorem.

Next, let me give a simple example. Suppose the offspring distribution is Poisson with parameter λ, so $X_i \sim \text{Pois}(\lambda)$. Then $X_1 + \cdots + X_t \sim \text{Pois}(t\lambda)$. Therefore, $\mathbb{P}(W = t) = \frac{1}{t}\mathbb{P}(X_1 + \cdots + X_t = t - 1) = \frac{1}{t}e^{-t\lambda}\frac{(t\lambda)^{t-1}}{(t-1)!}$, $t \geq 1$.

So the theorem lets us compute the distribution of total progeny by reducing it to a standard calculation for a sum of i.i.d. Poisson random variables.

Lemma 1

Let $(Z_n)_{n\geq 0}$ be a Galton–Watson branching process, where $Z_{n+1} = \sum_{i=1}^{Z_n} \xi_i^{(n)}$ is generation-size random variable, and let $W := \sum_{n=0}^{\infty} Z_n$ denote the total progeny. In the breadth-first exploration process, let exploration stopping time (aka extinction time of the BFS queue / active set) $\tau_0 := \inf\{t \geq 0 : \ A_t = 0\}$, where A_t is the number of active vertices after t exploration steps. Then $W = \tau_0$ (with the convention that both sides may be $+\infty$).

Remark. The size of the tree is exactly the number of BFS steps needed to finish exploring it. S_t is just the same BFS queue process written as a random walk becuse At each step, you remove one active vertex and add X_t new children, so the queue changes by $X_t - 1$.

Remark (Relation to Law of Total Progeny). When defined in terms of A_t, τ_0 is the extinction time of the active set, or the time when the BFS queue first empties. If defined in terms of S_n, it is the first hitting time of 0 for the walk S_t or the hitting time of 0 for the Lukasiewicz/random-walk exploration process. the first hitting time of 0 by the exploration random walk S_t (not arbitrary walk b/c it started at 1). This is a hitting time, not a return time, because the walk starts at 1, not 0. The lemma and the theorem are related, but they do different jobs. The first part of the theorem, $W = \tau_0$, is the same statement as the lemma, just written in the notation: S_t instead of A_t. The lemma defines $\tau_0 := \inf\{t \geq 0 : \ A_t = 0\}$, where A_t is the number of active vertices after t exploration steps; the theorem defines $\tau_0 := \inf\{t \geq 0 : \ S_t = 0\}$, where $S_t = 1 + \sum_{s=1}^{t} X_s - 1)$. But in the breadth-first exploration, S_t is exactly the active-set size A_t. So these are the same stopping time.

The lemma is an identification statement: $W = \tau_0$ almost surely, with the convention that both sides may be $+\infty$. Here, $W = \sum_{n=0}^{\infty} Z_n$ is the total progeny, while τ_0 is the first time the breadth-first exploration queue becomes empty. Thus the lemma tells us what random variable W is in terms of the exploration process.

By contrast, the theorem contains this same identification but goes further: it gives an explicit formula for the distribution of W. Namely, for every $t \geq 1$, $\mathbb{P}(W = t) = \mathbb{P}(\tau_0 = t) = \frac{1}{t}\,\mathbb{P}(X_1 + \cdots + X_t = t - 1)$.

So the lemma is structural, while the theorem is quantitative.

The lemma answers the question: *What is total progeny in the breadth-first exploration picture?* The answer is: $W = \tau_0$.

The theorem answers the stronger question: *What is the probability that the total progeny equals t?* The answer is: $\mathbb{P}(W = t) = \frac{1}{t}\,\mathbb{P}(X_1 + \cdots + X_t = t - 1)$.

From the lemma alone, one only gets $\mathbb{P}(W = t) = \mathbb{P}(\tau_0 = t)$, but this does not yet compute the value of that probability. To obtain the explicit formula, one must use the random-walk representation $S_t = 1 + \sum_{s=1}^{t}(X_s - 1)$ together with a ballot-theorem or cycle-lemma type hitting-time result.

Therefore, the theorem strictly strengthens the lemma: the lemma identifies the quantity, while the theorem computes its law.

Proof

Recall the breadth-first exploration partition at time t: $\mathcal{A}_t$ = active vertices, $\mathcal{E}_t$ = explored vertices, $\mathcal{N}_t$ = not-yet-seen vertices, and write their sizes as $A_t := |\mathcal{A}_t|$, $E_t := |\mathcal{E}_t|$, $N_t := |\mathcal{N}_t|$.

Step 1: One vertex is explored per unit time. By construction of the exploration, as long as there is at least one active vertex, the procedure takes the next active vertex in the queue and declares it explored. Therefore, for every $t \le \tau_0$, $E_t = t$.

Step 2: Partition identity. On the event $\{W < \infty\}$ (extinction), the Galton–Watson tree has exactly W vertices. At each time $t \le \tau_0$ every vertex is in exactly one of the three sets $\mathcal{A}_t, \mathcal{E}_t, \mathcal{N}_t$, so $A_t + E_t + N_t = W$, for all $t \le \tau_0$. Using $E_t = t$ from Step 1, we obtain $A_t + t + N_t = W$, for all $t \le \tau_0$.

Step 3: The case of non-extinction. If the process does not go extinct, then $W = +\infty$. In this case there are always more vertices to explore, so the active set never becomes empty, and hence $\tau_0 = +\infty$. Thus $W = \tau_0$ holds on $\{W = +\infty\}$.

Step 4: The case of extinction. Now suppose the process goes extinct and $W < \infty$. Write $W = n$. At time $t = \tau_0$, by definition of τ_0 the active set is empty:

$$A_{\tau_0} = 0.$$

Moreover, once the active set is empty, the exploration has terminated, so there are no unseen vertices remaining:

$$N_{\tau_0} = 0.$$

Plugging $t = \tau_0$ into the identity $A_t + t + N_t = W$ yields

$$0 + \tau_0 + 0 = n,$$

so $\tau_0 = n = W$. This proves the lemma.

Hitting-Time Theorem

Theorem (Sebastian Roch Theorem 6.2.5)**.** *Let $(R_t)_{t\geq 0}$ be a random walk started at 0 with i.i.d. increments $(U_t)_{t\geq 1}$ satisfying $\mathbb{P}(U_t \leq 1) = 1$. Fix a positive integer ℓ and let*

$$\sigma_\ell := \inf\{t \geq 0 : \ R_t = \ell\}.$$

Then, for every integer $t \geq 1$,

$$\mathbb{P}(\sigma_\ell = t) = \frac{\ell}{t}\,\mathbb{P}(R_t = \ell).$$

Remark. The event $\{R_t = \ell\}$ is a position event or endpoint event, and often called level-ℓ event at time ℓ, or "the walk visits ℓ at time t". But it does not by itself mean a hitting event. $\{R_t = \ell\}$ means the walk is at ℓ at time t but $\{\sigma_\ell = t\}$ means the walk hits ℓ for the first time at time t and σ_ℓ is the hitting time or first-passage time.

This theorem relates the first-passage event to the ordinary position event.

Remark (Intuition from simple random walk.). For the simple symmetric random walk, the increments satisfy $U_t \in \{-1, 1\}$, so in particular $\mathbb{P}(U_t \leq 1) = 1$. Thus the assumption

of the theorem is automatically satisfied. **The ballot theorem** implies that among all paths ending at ℓ, exactly a fraction ℓ/t of them avoid hitting ℓ before time t. Hence $\mathbb{P}(\sigma_\ell = t) = \frac{\ell}{t}\,\mathbb{P}(R_t = \ell)$. Thus the hitting-time theorem can be viewed as a generalization of the ballot theorem from simple random walk to random walks whose increments are bounded above by 1.

Law of Total Progeny

Theorem (Sebastian Roch Theorem 6.2.6)**.** *Let $(Z_n)_{n\geq 0}$ be a Galton–Watson branching process with offspring distribution ξ, and let W be its total progeny. In the breadth-first random-walk representation, let $(X_t)_{t\geq 1}$ be the i.i.d. offspring variables revealed by exploring vertices one by one, and define*

$$Y_t := X_t - 1, \qquad S_t := 1 + \sum_{s=1}^{t} Y_s \quad \textit{with } S_0 = 1, \qquad \tau_0 := \inf\{t \geq 0 : \ S_t = 0\}.$$

Then $W = \tau_0$, and for all $t \geq 1$,

$$\mathbb{P}(W = t) = \frac{1}{t}\,\mathbb{P}(X_1 + \cdots + X_t = t - 1).$$

Remark. Y_t = the increment of the exploration random walk (or simply the random walk increment).

X_t = the offspring variable of the t-th explored individual. When you explore one vertex, you remove one active vertex. You add X_t new children to the queue.

τ_0 = first hitting time of 0, or the first time the exploration walk hits 0. extinction time of the exploration process (in the branching-process interpretation)

$\tau = 0 = 1$ = immediate extinction, the root has no offspring. $P(\tau_0 = 1) = P(\varepsilon = 0)$

Proof. **Step 1: Identify W with the hitting time τ_0.**

In the breadth-first exploration, let A_t be the number of *active* vertices after t exploration steps. We start with one active vertex (the root), so $A_0 = 1$. At each time $t \geq 1$, if $A_{t-1} > 0$ we explore exactly one active vertex (so the active count decreases by 1) and we add its

offspring to the active set (so the active count increases by X_t). Hence, for $t < \tau_0$,

$$A_t = A_{t-1} - 1 + X_t.$$

With $Y_t := X_t - 1$ this becomes

$$A_t = A_{t-1} + Y_t, \qquad A_0 = 1,$$

so by iterating,

$$A_t = 1 + \sum_{s=1}^{t} Y_s = S_t \qquad \text{for all } t \leq \tau_0.$$

By definition, the exploration stops exactly when there are no active vertices left, i.e. when $A_t = 0$. Therefore

$$\tau_0 = \inf\{t \geq 0 : \ A_t = 0\} = \inf\{t \geq 0 : \ S_t = 0\}.$$

S_t is the breadth-first exploration process written as a random walk. It can be interpreted as: i) number of active (unexplored) vertices after t exploration steps; ii) the queue length in the BFS exploration; iii) the Lukasiewicz path / Lukasiewicz walk of the Galton–Watson tree.

A_t is combinatorial and offers a BFS viewpoint by emphasizing the tree exploration meaning (queue size): the number of active vertices after t steps. S_t emphasizes the random-walk (partial sum process) meaning: $S_t = 1 + \sum(X_s - 1)$. You can add A_t as concrete quantity in the tree and S_t as the same quantity after recognizing it has the form of a partial-sum process. But they are the same processes: queue process $\leftrightarrow$ random walk.

Now observe that *on extinction* (this means, "up to the extinction time") exactly one vertex is explored at each time $1, 2, \ldots, \tau_0$, and when the active set becomes empty the entire finite tree has been explored. Thus the number of explored vertices equals the total number of vertices in the tree, i.e. the total progeny W, and we get

$$W = \tau_0.$$

Law of total progeny

Step 2: Rewrite the event $\{W = t\}$ in terms of the sum $X_1 + \cdots + X_t$.

Since $S_t = 1 + \sum_{s=1}^{t}(X_s - 1)$, we have the identity

$$S_t = 1 + \big(X_1 + \cdots + X_t\big) - t.$$

Therefore

$$\{S_t = 0\} = \{X_1 + \cdots + X_t = t - 1\}$$

So

$$\tau_0 = \inf\{t \geq 0 : \ S_t = 0\} = \inf\{t \geq 0 : \ X_1 + \cdots + X_t = t - 1\}.$$

Step 3: Convert to a walk starting at 0 and apply Theorem 6.2.5 with $\ell = 1$.

Define a new process

$$R_t := 1 - S_t, \qquad t \geq 0.$$

Then $R_0 = 1 - S_0 = 0$, and for $t \geq 1$,

$$R_t - R_{t-1} = -(S_t - S_{t-1}) = -Y_t =: U_t.$$

Thus (R_t) is a random walk started at 0 with i.i.d. increments $U_t = -Y_t$.

Because $X_t \geq 0$ always, we have $Y_t = X_t - 1 \geq -1$, hence

$$U_t = -Y_t \leq 1,$$

so (R_t) satisfies the hypothesis $\mathbb{P}(U_t \leq 1) = 1$ in Theorem 6.2.5.

Next, note that

$$\tau_0 = \inf\{t \geq 0 : \ S_t = 0\} = \inf\{t \geq 0 : \ R_t = 1\}.$$

So τ_0 is exactly the first hitting time of level $\ell = 1$ by R:

$$\tau_0 = \sigma_1.$$

Applying Theorem 6.2.5 with $\ell = 1$ yields, for every $t \geq 1$,

$$\mathbb{P}(W = t) = \mathbb{P}(\tau_0 = t) = \mathbb{P}(\sigma_1 = t) = \frac{1}{t}\,\mathbb{P}(R_t = 1).$$

Finally, since $R_t = 1 - S_t$, we have $R_t = 1 \iff S_t = 0$, hence

$$\mathbb{P}(R_t = 1) = \mathbb{P}(S_t = 0) = \mathbb{P}(X_1 + \cdots + X_t = t - 1),$$

and therefore

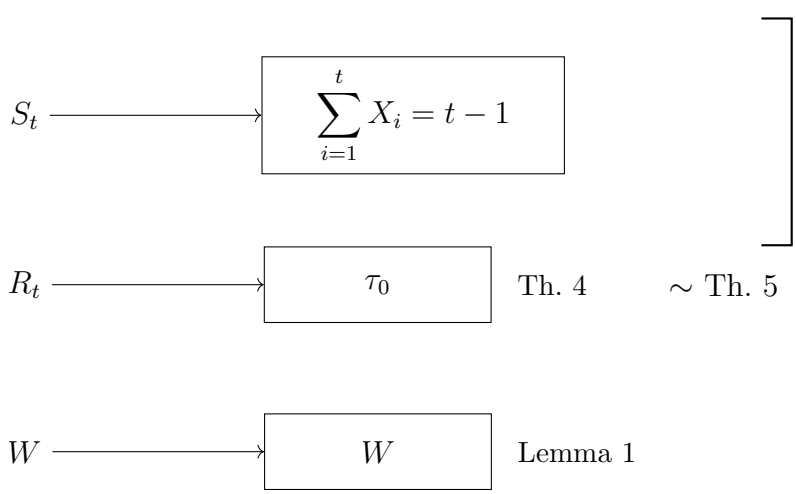
$$\mathbb{P}(W = t) = \frac{1}{t}\,\mathbb{P}(X_1 + \cdots + X_t = t - 1),$$

as claimed.

□

Remark. Note that total children = # edges = $t - 1$. This is because all vertices except the root has one parent. The beauty of this theorem is to convert W, which is a global tree quantity, into a sum of iid variables. W strongly depends on tree structure across all generations, but the sum of iid RVs with the same offspring law can be studied with standard tools, such as generating functions, binomial / Poisson / geometric formulas, local limit theorems, large deviation estimates, and random-walk methods.

Remark. Note that $\frac{1}{t}$ encodes the "first hitting time" information. This is the nontrivial part coming from the hitting-time theorem / ballot theorem.



Example 2

(Poisson branching process) Let $p_k = \frac{\lambda^k}{k!}e^{-\lambda}$, $\lambda > 0$. The the offspring law is Poisson(λ), which is a Galton–Watson branching process;the offspring counts of different individuals are i.i.d.; in the breadth-first exploration, the revealed variables $X_1, X_2, ...$ are iid with same law as ε, ie, Poisson(λ). Basically, **Example 2 is a Galton–Watson process with i.i.d. offspring law Poisson(λ) exactly the hypothesis of the theorem.** Then for every $t \geq 1$,

$$\mathbb{P}(W = t) = \frac{1}{t}\,\mathbb{P}(X_1 + \cdots + X_t = t - 1).$$

Since $X_1, \ldots, X_t$ are i.i.d. Poisson(λ), their sum is Poisson(λt). Therefore, $\mathbb{P}(X_1 + \cdots + X_t = t - 1) = \frac{(\lambda t)^{t-1}}{(t-1)!}e^{-\lambda t}$. Substituting into the formula for $\mathbb{P}(W = t)$ gives $\mathbb{P}(W = t) = \frac{1}{t} \cdot \frac{(\lambda t)^{t-1}}{(t-1)!}e^{-\lambda t} = \frac{(\lambda t)^{t-1}}{t!}e^{-\lambda t}$.

The distribution of τ_a

Let $a \in \mathbb{N}$ and let $(S_k)_{k\geq 0}$ be the simple symmetric random walk (SSRW) with $S_0 = 0$ and increments $S_k - S_{k-1} \in \{-1, +1\}$ i.i.d. with probability 1/2 each. Define the first hitting time of level a by

$$\tau_a := \inf\{k \geq 0 : \ S_k = a\},$$

and define the running maximum

$$S_n^* := \max_{k\leq n} S_k.$$

Then

$$\begin{aligned}\mathbb{P}(\tau_a \le n) &= \mathbb{P}(S_n^* \ge a) \\ &= \sum_{b\in\mathbb{Z}} \mathbb{P}(S_n^* \ge a,\ S_n = b) \\ &= \sum_{b\ge a} \mathbb{P}(S_n^* \ge a,\ S_n = b) \ + \ \sum_{b<a} \mathbb{P}(S_n^* \ge a,\ S_n = b).\end{aligned}$$

Step 1: the terms with $b \ge a$. If $b \ge a$, then $S_n = b$ forces $S_n^* \ge a$, hence

$$\mathbb{P}(S_n^* \ge a,\ S_n = b) = \mathbb{P}(S_n = b), \qquad b \ge a.$$

Therefore

$$\sum_{b\ge a} \mathbb{P}(S_n^* \ge a,\ S_n = b) = \sum_{b\ge a} \mathbb{P}(S_n = b) = \mathbb{P}(S_n \ge a).$$

Step 2: the terms with $b < a$ (reflection principle). Fix $b < a$ and consider paths that satisfy $S_n = b$ and $S_n^* \ge a$. Let θ be the *first* time the path hits a:

$$\theta := \inf\{k \le n :\ S_k = a\}.$$

Reflect the path after time θ across the horizontal line at height a:

$$\widetilde{S}_k := \begin{cases} S_k, & k \le \theta, \\ 2a - S_k, & k \ge \theta. \end{cases}$$

This mapping is a bijection between the set of paths with $(S_n^* \ge a,\ S_n = b)$ and the set of paths with endpoint $\widetilde{S}_n = 2a - b$. Since SSRW assigns equal probability to each length-n increment sequence, the bijection preserves probability. Hence

$$\mathbb{P}(S_n^* \ge a,\ S_n = b) = \mathbb{P}(S_n = 2a - b), \qquad b < a.$$

Consequently,

$$\sum_{b<a} \mathbb{P}(S_n^* \geq a,\ S_n = b) = \sum_{b<a} \mathbb{P}(S_n = 2a - b).$$

Step 3: rewrite the reflected sum. Let $c := 2a - b$. When $b < a$, we have $c > a$. As b runs over $\{\ldots, a-2, a-1\}$, the variable c runs over $\{a+1, a+2, \ldots\}$. Therefore

$$\sum_{b<a} \mathbb{P}(S_n = 2a - b) = \sum_{c>a} \mathbb{P}(S_n = c) = \mathbb{P}(S_n > a).$$

Putting Steps 1–3 together gives

$$\begin{aligned}\mathbb{P}(\tau_a \leq n) &= \mathbb{P}(S_n \geq a) + \mathbb{P}(S_n > a) \\ &= 2\,\mathbb{P}(S_n > a) + \mathbb{P}(S_n = a).\end{aligned}$$

By symmetry of SSRW, $\mathbb{P}(S_n > a) = \mathbb{P}(S_n < -a)$, hence

$$\mathbb{P}(\tau_a \leq n) = \mathbb{P}(S_n > a) + \mathbb{P}(S_n = a) + \mathbb{P}(S_n < -a) = \mathbb{P}\big(S_n \notin [-a, a-1]\big).$$

Step 4: obtain $\mathbb{P}(\tau_a = n)$. Using $\mathbb{P}(\tau_a = n) = \mathbb{P}(\tau_a \leq n) - \mathbb{P}(\tau_a \leq n-1)$, we get

$$\begin{aligned}\mathbb{P}(\tau_a = n) &= \mathbb{P}\big(S_n \notin [-a, a-1]\big) - \mathbb{P}\big(S_{n-1} \notin [-a, a-1]\big) \\ &= \mathbb{P}\big(S_n \notin [-a, a-1],\ S_{n-1} \in [-a, a-1]\big).\end{aligned}$$

From $S_{n-1} \in [-a, a-1]$ and a single ± 1 step, the only ways to exit $[-a, a-1]$ at time n are:

$$S_{n-1} = a - 1 \text{ and step } +1 \quad \text{or} \quad S_{n-1} = -a \text{ and step } -1.$$

Thus

$$\mathbb{P}(\tau_a = n) = \tfrac{1}{2}\,\mathbb{P}(S_{n-1} = a - 1) + \tfrac{1}{2}\,\mathbb{P}(S_{n-1} = -a).$$

By symmetry, $\mathbb{P}(S_{n-1} = -a) = \mathbb{P}(S_{n-1} = a)$, so

$$\mathbb{P}(\tau_a = n) = \tfrac{1}{2}\Big(\mathbb{P}(S_{n-1} = a - 1) + \mathbb{P}(S_{n-1} = a)\Big).$$

Finally, use the SSRW identity

$$\mathbb{P}(S_n = a) = \tfrac{1}{2}\,\mathbb{P}(S_{n-1} = a-1) + \tfrac{1}{2}\,\mathbb{P}(S_{n-1} = a+1),$$

to rewrite $\frac{1}{2}\,\mathbb{P}(S_{n-1} = a-1)$ as

$$\tfrac{1}{2}\,\mathbb{P}(S_{n-1} = a-1) = \mathbb{P}(S_n = a) - \tfrac{1}{2}\,\mathbb{P}(S_{n-1} = a+1).$$

Substituting and simplifying yields the boxed identity marked $(*)$ in the notes:

$$\boxed{\mathbb{P}(\tau_a = n) = \tfrac{1}{2}\Big(\mathbb{P}(S_{n-1} = a-1) - \mathbb{P}(S_{n-1} = a+1)\Big).} \qquad (*)$$

Exercise 8 (ballot/hitting-time identity)

Claim (Exercise 8). For SSRW and $a \in \mathbb{N}$,

$$\boxed{\mathbb{P}(\tau_a = n) = \frac{a}{n}\,\mathbb{P}(S_n = a)}, \qquad n \geq 1.$$

Why Exercise 8 is a special case of Theorem 6.2.5. To recover Exercise 8 from Theorem 6.2.5, take

$$R_t := S_t \quad \text{(the SSRW itself)}.$$

Then $R_0 = S_0 = 0$ and the increments are

$$U_t := R_t - R_{t-1} = S_t - S_{t-1} \in \{-1, +1\}.$$

In particular, $U_t \leq 1$ always, so the hypothesis $\mathbb{P}(U_t \leq 1) = 1$ holds. Now choose $\ell = a$. The

hitting time of level a for R is exactly τ_a:

$$\sigma_a = \inf\{t \geq 0 : \ R_t = a\} = \inf\{t \geq 0 : \ S_t = a\} = \tau_a.$$

Applying Theorem 6.2.5 with $\ell = a$ and $t = n$ gives

$$\mathbb{P}(\tau_a = n) = \mathbb{P}(\sigma_a = n) = \frac{a}{n}\,\mathbb{P}(R_n = a) = \frac{a}{n}\,\mathbb{P}(S_n = a),$$

which is exactly the statement of Exercise 8.

Remark (relation to $(*)$). Identity $(*)$ can be viewed as an alternative route to the same result: once one rewrites $\mathbb{P}(S_{n-1} = a - 1) - \mathbb{P}(S_{n-1} = a + 1)$ in closed form (e.g. via binomial coefficients) and simplifies, one obtains $\mathbb{P}(\tau_a = n) = \frac{a}{n}\mathbb{P}(S_n = a)$, which is the ballot/hitting-time formula.